

Computer Science Capstone Topic Approval Form

The purpose of this document is to help you clearly explain your capstone topic, project scope, and timeline. Identify each of these areas so that you will have a complete and realistic overview of your project. Your instructor cannot sign off on your project topic without this information.

Note: You must fill out and submit this form. Space beneath each number will expand as needed.

Note: Any costs associated with developing the application will be the responsibility of the student.

INFORM INSTRUCTOR:

Potential use of proprietary company information: (Y/N) N

ANALYSIS:

1. Project topic and description:

My client ABC Home Loans needs a tool to speed up their mortgage loan approval process by filtering out applications with potential for approval and which do not.

2. Project purpose and goals:

To provide a Mortgage Application Analysis tool that the Loan Officers can use to invest time into applicants who are likely to get a loan approved.

3. Descriptive method:

A variety of different visualization will be used such as a Barplot to show which value indicates the highest chance of success. Also a Heatmap will be a great visualization to compare each criteria with another to see the impact they have on the loan approval. Also a Catplot to compare categories with one another to see which is likelier to get approved.

4. Predictive or prescriptive method:

Using classification models, the goal will be to show the best accuracy for loan approval using: RandomForestClassifier, KNeighborsClassifier, Support Vector Classifier (SVC), and Logistics Regression.

DESIGN and DEVELOPMENT:

1. Computer science application type (select one):

- Mobile (indicate Apple or Android)



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- Web
 - Stand-alone
2. Programming/development language(s) you will use:
Python
 3. Operating system(s) or platform(s) you will use:
Windows 10
 4. Database Management System you will use:
N/A
 5. Estimated number of hours for the following:
 - i. Planning and design:20
 - ii. Development:30
 - iii. Documentation:15
 - iv. Total:65
 6. Projected completion date:1/20/2025

IMPLEMENTATION and EVALUATION:

1. Describe how you will approach the execution of your project.
 1. All requirements will be established first with the clients.
 2. Application will be built piece by piece to ensure everything is working before jumping to the next step.
 3. The data will be used to test the different non-descriptive models that were proposed to see which comes out on top for accuracy.
 4. Once operational, the documentation will be written and the package will be delivered to the client.
- ☐ **This project does not involve human subjects research and is exempt from WGU IRB review.**

STUDENT'S SIGNATURE

By signing and submitting this form, you acknowledge that any costs associated with the development and execution of the application will be your (the student's) responsibility.

INSTRUCTOR'S SIGNATURE:

INSTRUCTOR APPROVAL DATE:

